

Enhancing Emotional Support Chatbots with Machine Learning and Linguistic Analysis

Khethavath Sunil Naik
khethavathn@iitbhillai.ac.in
Indian Institute of Technology Bhilai
Data Science and Artificial Intelligence
Bhilai, Chhattisgarh, India

Abstract

Conversations about mental well-being surface in the most ordinary settings now. A friend might message late at night saying they feel strangely unsettled, or someone may confess during a walk that they have been carrying a quiet worry for weeks. Despite how common these moments have become, many people still do not feel comfortable seeking formal help, and others simply do not have access to it. That gap has encouraged the rise of emotional support chatbots. They are always awake, they never seem impatient, and they offer a sense of companionship, although a fragile one.

Yet most of these systems respond in ways that feel oddly flat. They often sound grammatically correct but emotionally hollow, as if the model understood the shape of the sentence but not the feeling inside it. This project grew out of the suspicion that emotional nuance is being lost somewhere between classification and generation.

EmoCare attempts to pay closer attention. It uses a fine-grained BERT classifier trained on thirty-two emotional categories, which include not just broad moods like joy or anger, but also more subtle states such as embarrassment, nostalgia, or apprehension. The classifier is strengthened through a semi-supervised loop that encourages it to learn from its own confident predictions. For response generation, I rely on a compact transformer running through the Ollama framework, which seemed like a practical compromise between flexibility and computational modesty. Instead of overwhelming verbosity, the goal was to produce messages that feel considerate and aware of the user's emotional direction.

To understand how supportive these responses truly are, I examine them through a psycholinguistic lens influenced by LIWC. This includes looking at affective markers, social-language cues, and small shifts in sentiment across a conversation. When compared with a template-based baseline, EmoCare offers noticeably warmer and more grounded interactions. While it remains far from replacing human care, the system hints at how small adjustments in emotional modeling can create conversations that feel less mechanical and more genuinely attentive.

Keywords

Emotion Recognition, Empathetic Dialogue, GPT2 generation, LLaMA Generation, BERT Classification, Linguistic Inquiry and Word Count (LIWC), Semi-Supervised Learning, NLP for Mental Health

ACM Reference Format:

Khethavath Sunil Naik. 2025. Enhancing Emotional Support Chatbots with Machine Learning and Linguistic Analysis. In *Proceedings of Machine Learning Conference 2025 (ML'25)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/XXXXXXX.XXXXXXX>

1 Introduction

People rarely express emotions cleanly. Someone might laugh while saying they are anxious about an upcoming interview. Another person may admit they feel excited about a big opportunity, although a bit of fear creeps in as well. These moments are familiar, and they remind us that emotional communication is rarely a straight line. Yet when individuals do look for a place to share these feelings, especially those who do not want to burden family or friends, they often end up turning to digital platforms. Emotional support chatbots have become one of those quiet companions. They offer a space where nobody judges or interrupts, which can be surprisingly comforting.

Still, many of these systems respond in ways that feel stiff. Anyone who has interacted with a typical chatbot will recognize the pattern. The responses are polite, occasionally sympathetic, but they rarely match the emotional weight of what the user says. A message like "I feel guilty about letting someone down" might receive a generic "I'm sorry to hear that" response, which feels more like a formality than comfort. This mismatch is not trivial. When someone is reaching out, even tentatively, the tone of the reply matters.

This project tries to see how far we can push emotional sensitivity using current machine learning tools. Rather than treating emotion as a two- or three-level scale, I work with a dataset containing thirty-two distinct emotional states. Some are obvious, such as joy or anger, but others capture niche but emotionally meaningful categories like "sentimental" or "proud" or "embarrassed." Working with these finer labels forces the classifier to pick up details hidden in phrasing and context.

But recognizing emotion is only part of the story. A chatbot that identifies sadness but responds awkwardly is not helpful. To address this, I integrate a small transformer model through the GPT2 and Ollama ecosystem. It does not aim to mimic large-scale commercial models. Instead, it is used as a calm and controlled generator, capable of responding with language that feels more naturally paced.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted by ACM, provided that the copies are not made for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

ML'25, IIT Bhilai, India
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-XXXX-X/25/09
<https://doi.org/10.1145/XXXXXXX.XXXXXXX>

Evaluating these systems requires more than accuracy and BLEU scores. Those metrics are useful but incomplete. If we want to understand emotional support, we need to ask how the responses sound to a human ear. Do they acknowledge the user's feelings? Do they show a small attempt to connect? Tools inspired by LIWC help answer these questions by analyzing psychological and social cues in language. With this approach, the goal is not to produce a perfect conversational partner but to create one that feels less superficial.

1.1 Background and Motivation

Mental health conversations have always existed, but they appear in different forms now. Someone might share a vulnerable thought over a quick text message or post anonymously online hoping a stranger might understand. This increase in openness does not necessarily mean people feel supported. Many hesitate to reach out to professionals because therapy can feel intimidating or financially out of reach. In some communities, the stigma around mental health remains strong, and individuals avoid seeking help altogether.

This is where chatbots often enter the picture. They do not replace human presence, yet they offer a private space to begin sorting through one's feelings. When used thoughtfully, they may serve as stepping stones toward more structured support. However, their limitations are visible. Many fall back on generic templates or recycled phrasing. Some avoid acknowledging complex emotions altogether, preferring safe but shallow replies. These issues motivated the design of EmoCare, which aims to treat emotional nuance with a bit more care.

1.2 Problem Statement

Two gaps repeatedly appear when examining existing emotional-support chatbots. First, the emotional classifications they rely on are often too coarse. Trying to describe all human emotion using only three categories feels blunt. People communicate subtle states, and systems should be able to recognize them if they hope to respond meaningfully.

Second, generating supportive responses is harder than it appears. Fluency does not guarantee empathy. A model can craft well-structured sentences yet miss the tone entirely. As a result, the user may feel as though they are speaking into a polished but indifferent machine.

This project addresses these issues by focusing on:

- building a fine-grained emotion classifier capable of recognizing nuanced affect, and
- generating responses that align with those emotions in a gentle and context-aware way.

1.3 Objectives

The primary objectives include:

- (1) Training a BERT-based model to classify thirty-two emotional states with enough sensitivity to handle subtle expressions.
- (2) Using semi-supervised learning to strengthen classification, especially for rare labels.
- (3) Incorporating a compact LLaMA-style model through Ollama to generate responses that adapt to emotional cues.

- (4) Applying psycholinguistic analysis to evaluate how empathetic, warm, or socially attentive the generated responses are.
- (5) Exploring sentiment shifts and conversation-level emotional dynamics to understand how interactions evolve over time.

1.4 Scope and Delimitations

This project focuses solely on text-based dialogue. Visual or auditory signals are not included, although they could expand emotional reasoning in future work. The system is not designed to offer clinical guidance. It does not attempt to diagnose mental health conditions. Instead, it tries to create conversations that feel a little less mechanical and more understanding. The generative model is intentionally lightweight to keep the system interpretable and manageable.

1.5 Novelty and Contributions

The contributions of this work include:

- A fine-grained emotional classifier strengthened by semi-supervised augmentation.
- A controlled generative system using a GPT2, LLaMA-style transformer integrated through Ollama.
- A multi-layered evaluation combining computational metrics with psycholinguistic measures.
- A discussion of emotional tone, conversational warmth, and the boundaries of automated support.

1.6 Organization of the Report

The remainder of the report is divided as follows. Section 2 explores related research such as emotion classification methods, generative dialogue systems, and the role of psycholinguistic cues. Section 3 explains the dataset and preprocessing strategy. Sections 4, 5, and 6 discuss the classical models, the BERT classifier, and the semi-supervised framework. Section 7 focuses on the generative model. Section 8 introduces linguistic and sentiment analysis. Section 9 provides results and reflections. Section 10 concludes the work.

2 Literature Review

Research on emotionally aware conversational systems sits at the intersection of natural language processing, psychology, and human-computer interaction. It is a space shaped by equal parts hope and skepticism. People want technology that feels supportive, but there is understandable hesitation about letting a machine talk about emotions at all. This chapter looks at how previous work has approached the problem, including its obvious strengths and the quieter limitations that often go unnoticed.

2.1 Mental Health and Technology

The idea of using technology for emotional support appeared long before modern neural networks. Early systems like ELIZA surprised people by reflecting their own words back at them. While simple, it revealed something interesting: people sometimes project warmth onto even the most mechanical text when they feel unheard elsewhere. Since then, digital mental-health interventions have expanded dramatically. Mobile therapy apps encourage journaling or breathing exercises, and online peer-support platforms host anonymous discussions about personal struggles.

Despite this progress, emotional support systems remain imbalanced across regions. Rural communities in India, for instance, often lack even basic mental-health infrastructure. Several studies point to the convenience of digital tools for those who hesitate to speak openly with family. However, access does not guarantee emotional depth. Many systems feel transactional. They respond quickly but often miss the tone or intention behind a message. These gaps are reminders that emotional modeling is not simply a machine-learning problem but one that requires cultural awareness and psycholinguistic nuance.

2.2 Emotion Recognition and Classification

Emotion classification is usually described as a technical task, although it rarely feels that simple when reading actual text written by people. Traditional approaches relied on lexicons. If a sentence contained words like “angry,” “furious,” or “annoyed,” the system labeled the emotion accordingly. These methods worked reasonably well for clear cases, but real emotional expression is often indirect. Someone may say, “I cannot believe that happened,” which could imply shock, guilt, or frustration depending on the situation.

With the rise of neural networks, classification improved. Recurrent models, especially LSTMs, captured sequences more effectively. However, transformer-based models transformed the field. BERT in particular showed an uncanny ability to notice subtle patterns in phrasing and context. Instead of focusing on keywords alone, it learns from how words relate to one another across the entire sentence. This capability is especially useful when dealing with emotions like embarrassment or pride, which rarely appear through explicit keywords.

Yet even with strong models, classification becomes harder when the number of emotional categories increases. Researchers often use six or eight labels, which simplifies training but loses nuance. Fine-grained classification, such as the thirty-two emotions used in this work, has been explored far less. It offers richer insight but introduces challenges with data imbalance and semantic overlap. For example, the difference between “afraid” and “terrified” may not be consistently expressed across individuals. This project works within that complexity rather than reducing it.

2.3 Dialogue Systems and Empathetic Response Generation

The evolution of dialogue systems is well documented. Early bots followed strict templates. Later, sequence-to-sequence models produced smoother responses, although they often fell into repetitive patterns. The push toward transformer-based generators opened a new chapter. Large language models can craft impressively fluid responses, but fluency alone is not a measure of emotional sensitivity.

Empathy-focused dialogue models attempt to adjust for this. Some researchers incorporate emotion labels directly into the generation process. Others train models on supportive conversations from counseling datasets. The major issue is that even sophisticated models can sound detached. They know how to continue a conversation but sometimes fail to grasp the weight behind the user’s words.

A smaller, controlled model, like the LLaMA-style generator used in this project through the Ollama framework, offers a different direction. While it may not produce long monologues, its responses can be shaped intentionally. With careful prompting, it can stay closer to the emotional logic of the conversation instead of drifting into unrelated or generic phrasing.

2.4 Linguistic and Psycholinguistic Analysis

Psycholinguistics offers a useful lens for evaluating supportive dialogue. Tools such as LIWC break language into psychological categories: affective processes, social references, cognitive mechanisms, and so on. Although the tool is often used for analyzing essays or therapy transcripts, the same ideas apply to chatbot responses. For instance, a supportive reply often reflects the user’s feelings (“It sounds like this has been weighing on you”), uses social markers (“you,” “we”), and expresses some form of understanding.

Several studies show correlations between LIWC categories and perceived empathy. High emotional resonance often involves acknowledging uncertainty, using warm language, or validating the user’s experience. These patterns informed the evaluation strategy of EmoCare. Rather than focusing exclusively on model accuracy, psycholinguistic cues help illustrate whether the system’s responses feel emotionally grounded.

2.5 Semi-Supervised Learning and Data Scarcity

Emotion datasets are notoriously imbalanced. Some emotions appear constantly—sadness, happiness, anger—while others show up only sporadically. Traditional supervised training often struggles with these rare classes, causing models to default to common labels. Semi-supervised learning is a practical workaround. By letting the classifier generate labels for unlabeled text and retraining with high-confidence predictions, the model gradually improves its sensitivity to subtle or infrequent states.

Self-training approaches have successfully improved performance in several NLP tasks. For emotion recognition, the method helps amplify patterns that are too rare for supervised learning alone. The semi-supervised loop used in this project follows this logic: let the model guide its own learning where appropriate, while maintaining control through confidence thresholds.

2.6 Research Gaps

The literature highlights two recurring gaps. First, there is relatively little work combining fine-grained emotional classification with generative systems in a controlled, lightweight manner. Second, evaluation often relies on computational metrics that do not capture how people feel when reading the responses. These gaps motivated the multi-layered approach used in EmoCare.

3 Dataset and Preprocessing

Working with emotional text requires patience. People express feelings in roundabout ways, occasionally contradict themselves, and sometimes switch tones mid-sentence. The dataset chosen for this project captures some of this complexity, though it is still a simplified version of how people talk in real life.

3.1 Dataset Selection

The Empathetic Dialogues dataset was selected because it reflects everyday emotional conversations rather than curated counseling transcripts. Each dialogue is anchored around a specific emotion chosen by the speaker. For example, a user might discuss losing an old photo album (nostalgia) or the frustration of being overlooked at work (jealousy). These scenarios are relatable, and they reveal how nuanced emotional expression can be.

Unlike sentiment datasets that classify text as positive, negative or neutral, Empathetic Dialogues provides thirty-two emotion categories. This diversity is both a strength and a challenge. It encourages fine-grained modeling but introduces heavy class imbalance. Some emotions appear thousands of times, while others show up only a few hundred times.

3.2 Structure of the Dataset

The dataset consists of short multi-turn conversations. Each dialogue begins with the speaker explaining a situation and continues with a listener responding in a supportive manner. Because the original dataset was designed for empathy research, the conversational tone tends to be gentle and understanding. This structure is useful for both classification and generation tasks.

Each entry includes:

- the user's context or emotional disclosure,
- the listener's supportive response,
- a labeled emotional category.

This makes the dataset well suited for pairing emotion recognition with response modeling.

3.3 Preprocessing Pipeline

Before training, the text goes through several steps. Although preprocessing can be tedious, small details often determine how well a model learns.

3.3.1 Text Cleaning. Text is lowercased for consistency, although some emotion cues in uppercase messages (like "I CAN'T BELIEVE THIS") are naturally lost. URLs and unnecessary tokens are removed. I intentionally avoid aggressive cleaning because emotional phrasing sometimes relies on imperfect grammar, hesitation marks like "umm," or expressive punctuation such as ellipses.

3.3.2 Tokenization and Lemmatization. Tokenization is done using spaCy. Lemmatization helps reduce sparsity in classical models but is avoided for BERT since its tokenizer handles subword patterns effectively. Emotional expressions like "scared," "scary," or "scaring" map to similar embeddings without needing explicit lemmatization.

3.3.3 Emotion Label Encoding. Each of the thirty-two emotion categories is mapped to an integer. This step seems trivial, but ordering matters during visualization. Grouping semantically similar emotions together later helps interpret confusion matrices more meaningfully.

3.3.4 Sentiment Extraction. Although this project focuses on emotion categories, sentiment still provides useful information. Tools like TextBlob or VADER produce polarity scores that help track whether conversations drift toward positivity or negativity over time.

3.4 Data Quality Analysis

Emotion datasets usually show similar patterns. People talk more about sadness, worry, or joy, while states like "faithful" or "content" appear less frequently. The Empathetic Dialogues dataset is no exception. Some categories are heavily skewed. A model trained naively would overpredict common emotions.

To address this, I examine distribution charts and text length patterns. Shorter inputs sometimes convey strong emotions ("I feel awful today"), while longer ones often express mixed or uncertain emotional states. This complexity informs how the classifier and generator should be trained.

3.5 Preparation for Models

3.5.1 Classical Models. For Logistic Regression, Random Forests, and SVMs, TF-IDF vectors are used. Unigrams and bigrams capture common phrases. Although these models lack deep contextual understanding, they serve as useful baselines.

3.5.2 BERT Classifier. The BERT tokenizer splits sentences into subwords. Context length is adjusted to keep most inputs intact without excessive padding. Since BERT is sensitive to subtle context shifts, preserving punctuation helps.

3.5.3 Generative Model Preparation. For the LLaMA-style generator, each training sample is formatted as a structured prompt:

```
[Emotion: X] [User: ...] [Response: ...]
```

This format encourages the model to rely on emotional cues while responding.

3.6 Storage and Versioning

Version control is maintained using Git for scripts and preprocessing logs. Dataset variations (such as pseudo-labeled extensions) are stored separately to avoid confusion.

This careful handling ensures reproducibility, which is vital when dealing with semi-supervised methods where results can drift depending on how confidence thresholds are tuned.

4 Classical Machine Learning Models

Before diving into larger transformer-based models, I wanted to understand how far traditional machine learning techniques could go with a dataset centered around emotional expression. Older models still have a certain charm in their simplicity. They force you to think more carefully about features instead of relying on enormous pretrained representations. Moreover, comparing them with modern models helps show what is gained and what is lost along the way.

4.1 Introduction

Emotion classification often starts with the assumption that text can be represented by a bag of words or, slightly more refined, TF-IDF features. This is not as outdated as it sounds. Many emotional cues are contained in common lexical patterns. Someone saying "I feel awful" or "I cannot stop crying" is easy enough for simple models to catch. The trouble appears when phrasing becomes indirect or when the emotional meaning depends on context rather than vocabulary.

Still, traditional models act as a good benchmark. Even if they are not perfect, they reveal whether emotional signals are strong enough to be picked up without deep contextual embedding.

4.2 Feature Representation with TF-IDF

Texts were vectorized using TF-IDF with unigrams and bigrams. I avoided trigrams since the dimensionality becomes unwieldy and rarely improves performance for conversational data. Certain bigrams, such as “so sorry” or “felt embarrassed,” turned out to be surprisingly strong emotional indicators.

Stopword removal was kept minimal. Words like “I,” “you,” or “feel” often hold emotional importance in dialogue, and removing them makes the text unnaturally sparse.

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \times \log\left(\frac{N}{\text{df}(t)}\right)$$

$\text{tf}(t, d)$ = number of occurrences of term t in d

$$\text{idf}(t) = \log\left(\frac{N}{\text{df}(t)}\right)$$

4.3 Logistic Regression

Logistic Regression tends to be underestimated. Despite its simplicity, it handles high-dimensional sparse text fairly well.

4.3.1 Performance and Observations. It performed better than I initially expected, especially for emotions with distinct lexical patterns like “angry,” “sad,” or “joyful.” However, subtle emotions like “content,” “proud,” or “nostalgic” proved difficult. Logistic Regression seems to rely heavily on obvious keywords, and in many dialogues, emotional meaning is conveyed indirectly.

One interesting observation was that it confused “anxious” and “worried” frequently, which suggests that lexical overlap between those categories is high. In real conversations, people often blur these emotions themselves, so perhaps the model is not entirely wrong.

$$\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

4.4 Random Forest

Random Forests struggled more than Logistic Regression. The dataset is too sparse and too high-dimensional for tree-based models to thrive.

4.4.1 Performance. The model tended to overfit quickly, memorizing rare word combinations. It also produced inconsistent results across runs due to the randomization in tree construction. Emotion categories with few samples suffered the most. Trees prefer clear thresholds, and emotional language rarely fits into predictable patterns.

4.5 Support Vector Machines

SVMs with linear kernels performed reasonably well, almost on par with Logistic Regression. They captured some subtle distinctions

that Random Forests missed, although training time was noticeably longer.

4.5.1 Performance Characteristics. Certain emotions such as “jealous” or “proud” remained difficult. These states appear infrequently, and the language used to describe them varies widely across individuals. A person may describe pride indirectly through storytelling rather than explicit self-congratulation, which is challenging for TF-IDF based models.

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$

$$\text{SVM objective: } \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$$

4.6 Comparative Discussion

Among classical models, Logistic Regression offered the most stable performance. SVMs provided slightly higher precision for a few emotions but required more careful tuning. Random Forests, surprisingly, were the weakest performers. These results reinforced the idea that high-dimensional emotional text requires representation learning beyond TF-IDF.

While traditional models cannot capture contextual nuance, they do highlight where emotional cues are strongest and where they become ambiguous. These insights became helpful later when examining BERT’s confusion patterns.

5 BERT-Based Emotion Classifier

After working with classical baselines, it became clear that handling thirty-two emotional categories requires a model capable of interpreting context, subtle expressions, and the relational structure between words. BERT fits naturally into this role.

5.1 Why Use BERT?

BERT’s ability to look both forward and backward in a sentence allows it to interpret phrases whose emotional meaning depends on surrounding context. Consider:

“I finally talked to my parents. It went better than I expected.”

Emotion here is not explicitly stated. A keyword-based model might fail to detect the sense of relief or cautious optimism. BERT, however, can capture emotional intent because it understands how “better than I expected” modifies the preceding sentence.

Moreover, fine-grained emotions often rely on tone. For example, embarrassment may be expressed indirectly, through hesitations or soft qualifiers. Phrases such as “I guess I messed up” carry emotional weight that BERT is better suited to interpret than TF-IDF models.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

5.2 Model Architecture

The model uses bert-base-uncased with a classification head that maps the pooled output to thirty-two emotion labels. A dropout

layer of 0.3 helps control overfitting. The architecture is compact enough to train efficiently on a single GPU.

$$\hat{y} = \text{softmax}(Wh_{\text{CLS}} + b)$$

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

5.3 Training Setup

Training was done for three epochs with a learning rate of 2×10^{-5} . Batch size was set to 16. Longer training led to overfitting for some rare classes, which is unsurprising given their small sample sizes.

During training, I noticed that some emotions converged faster than others. Common categories such as “sad,” “afraid,” or “angry” reached stable performance quickly. Rare ones like “faithful” or “prepared” fluctuated more across epochs.

$$\mathcal{L}_{CE} = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

5.4 Performance

The supervised BERT model achieved around 67 percent accuracy. This number may seem modest compared to typical BERT classification tasks, but fine-grained emotional classification is considerably harder. Many categories overlap semantically or appear only in specific conversational contexts.

For example:

- “Hopeful” and “optimistic” often blur together.
- “Anxious” and “afraid” shade into each other depending on word choice.
- “Embarrassed” is expressed inconsistently across individuals.

Even humans struggle to categorize certain sentences consistently. Emotional labels are inherently fuzzy.

5.5 Confusion Patterns

A few patterns stood out:

- BERT often confused “surprised” with “excited,” which makes sense given their emotional closeness.
- “Jealous” was frequently misclassified as “annoyed,” possibly because both involve frustration.
- “Sentimental” and “nostalgic” were interchangeable in many instances.

These confusions are not necessarily errors but reflections of how people naturally talk about emotions.

6 Semi-Supervised Learning with Pseudo-Labeling

While the BERT classifier performed reasonably well, rare emotional states remained a challenge. Some categories had fewer than a few hundred samples. Instead of collecting new data, which is difficult for emotional text, I turned to semi-supervised learning.

6.1 Motivation

Pseudo-labeling is an interesting approach because it acknowledges something subtle: a model that is moderately good at classification can help improve itself by labeling unlabeled data it feels confident

about. Emotion recognition is messy, but high-confidence predictions often reflect meaningful patterns.

The hope is not to blindly trust the model but to expand the training dataset in ways that reinforce underrepresented categories.

6.2 Creating the Unlabeled Set

A subset of the validation data was treated as unlabeled. This might sound counterintuitive, but validation text still contains emotionally rich expressions. Using it as unlabeled data allows the model to extract additional patterns without the need for new annotations.

6.3 Generating Pseudo-Labels

The trained BERT model predicts emotion labels and confidence scores. Only predictions above an 85 percent threshold are kept. This threshold is intentionally high to limit noise.

Interestingly, a substantial portion of the pseudo-labeled data came from emotions like “hopeful,” “disappointed,” and “grateful,” which tend to have clear linguistic cues.

$$\tilde{y} = \begin{cases} \arg \max_k p_\theta(y = k | x) & \text{if } \max_k p_\theta(y = k | x) \geq \tau \\ \text{ignore sample} & \text{otherwise} \end{cases}$$

$$D_{\text{aug}} = D_{\text{labeled}} \cup \{(x_i, \tilde{y}_i)\}$$

6.4 Retraining with Augmented Data

After adding the pseudo-labeled samples, BERT was retrained from scratch. The augmented dataset improved accuracy to around 72 percent. More importantly, rare classes saw the largest gains. Categories such as “apprehensive” and “sentimental” improved by several percentage points.

$$\mathcal{L} = \mathcal{L}_{\text{supervised}} + \lambda \mathcal{L}_{\text{pseudo}}$$

6.5 Analysis

The semi-supervised loop appears to stabilize emotional boundaries. Certain categories that were previously ambiguous became more distinct. While the approach is not flawless—if the initial model is biased, pseudo-labeling may reinforce biases—it nonetheless provides a practical way to strengthen emotional modeling with limited labeled data.

7 Response Generation with a GPT2 Generator via Hugging Face and LLaMA-Style Generator via Ollama

Detecting emotion is one thing; responding to it is another entirely. A supportive reply must do more than mirror the user’s words. It should acknowledge feelings, express understanding, and guide the conversation gently.

For generation, I used a compact LLaMA-style transformer model running locally through the Ollama framework. This choice was partly practical. Large language models, while powerful, tend to generate verbose or overly polished responses that feel unnatural in emotional contexts. A smaller model is easier to steer and less prone to wandering.

7.1 Why a Smaller Model?

I noticed that large models sometimes “over-perform” emotionally. They attempt to sound wise or overly therapeutic, which can come across as artificial or even intrusive. Smaller models produce shorter, more grounded responses, which align better with the project’s goals.

7.2 Prompt Structure

Each generation prompt followed a consistent structure:

```
Emotion: <predicted emotion>
User: <user message>
Respond in a supportive and understanding
tone.
```

This formatting keeps the model anchored. It is not asked to solve emotional problems or offer advice, only to respond with a sense of awareness.

7.3 Style of Generated Responses

The responses tended to be:

- short and measured,
- mildly conversational,
- emotionally aligned with the user’s tone,
- and often uncertain in a very human way.

For instance, given a message about feeling nervous, the model might reply:

```
“It makes sense that you are feeling uneasy. Maybe
the situation means a lot to you. If you want to talk
through it, I am here.”
```

There is a gentle hesitancy in phrasing, which feels more authentic than confident declarations.

7.4 Limitations

The model sometimes repeats generic phrases. It also struggles with deeply complex emotional disclosures, occasionally leaning on safe but vague expressions. That said, these issues are present even in larger state-of-the-art models and reflect the ongoing difficulty of generating supportive language responsibly.

8 Evaluation and Analysis

Evaluating an emotionally supportive chatbot is a difficult task because the qualities we care about most are also the hardest to measure. Accuracy and F1 scores tell part of the story, but they do not capture warmth or tone. A response might be technically correct yet feel emotionally off. For this reason, the evaluation process combines traditional metrics with more qualitative, psychologically grounded analyses.

8.1 Classification Metrics

The supervised BERT classifier achieved an accuracy of about 67 percent, improving to roughly 72 percent through the semi-supervised loop. These numbers should be interpreted cautiously. Emotional labeling is not as crisp as categorizing spam email or detecting sentiment polarity. Many emotions blur into one another, and the dataset reflects the natural inconsistency of human expression.

Weighted F1 scores remained in a similar range. Macro F1 was slightly lower because rare emotions still pose challenges. Yet the semi-supervised method helped enough to make the classifier feel more stable, particularly for categories that earlier models frequently ignored.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

8.2 Confusion Matrix Interpretation

The confusion matrix reveals interesting social patterns. Emotions like “jealous,” “embarrassed,” and “sentimental” show substantial overlap with neighboring categories. In contrast, broader states such as “sad” or “afraid” are classified accurately with little confusion.

One pattern I found amusing is that “nostalgic” and “sentimental” were confused at least as often as they were predicted correctly. When you read the actual messages, the distinction barely matters. People talk about old memories with a mix of fondness and melancholy, and the boundary between the two is thin.

8.3 Top-k Accuracy

Top-3 accuracy reached well over 90 percent. This suggests that even when the classifier misses the exact label, its alternatives are usually reasonable. From a conversational perspective, this is encouraging. If the model internally weighs several plausible emotions, the generator can still craft a response that feels appropriately attuned.

8.4 Evaluation of Generated Responses

Assessing generated text is notoriously subjective. Standard metrics like BLEU or ROUGE are helpful but limited. A supportive reply does not necessarily match a reference response word for word. Instead, I interpret these metrics as indicators of linguistic coherence rather than emotional skill.

8.4.1 BLEU and ROUGE. BLEU-1 and ROUGE-1 scores showed moderate improvements over a template-based baseline. While not spectacular, the gains suggest the generator stays closer to conversational patterns present in the dataset.

Longer BLEU metrics drop as expected, which is normal when responses are short and shaped more by tone than by structural matching.

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

$$\text{ROUGE-1} = \frac{\text{Overlapping Unigrams}}{\text{Total Unigrams in Reference}}$$

8.4.2 Semantic Similarity. BERTScore captured semantic alignment more effectively. Generated responses closely mirrored the meaning and emotional direction of reference responses, even when phrasing varied substantially.

This reflects a key design goal: consistency with emotional tone, not textual mimicry.

$$\text{BERTScore}(x, y) = \frac{1}{|x|} \sum_i \max_j \cos(e(x_i), e(y_j))$$

8.5 Empathy and Psycholinguistic Measures

To understand whether the chatbot genuinely sounds supportive, I applied psycholinguistic analysis informed by LIWC. This involved examining:

- affective words (e.g., “worried,” “relieved,” “upset”),
- social language (“you,” “we,” direct references),
- cognitive processing words (“maybe,” “I guess,” “it seems”),
- and emotional acknowledgments (“I can see why that would bother you”).

8.5.1 Findings. Generated responses show noticeably higher use of emotional-reflection words compared to the baseline. They also use more hedging language. Interestingly, the presence of slight uncertainty (“maybe,” “it sounds like,” “I’m not entirely sure, but...”) makes the replies feel more human rather than mechanical.

This hesitancy mirrors how supportive friends often speak. It avoids overconfidence and leaves space for the user to elaborate.

8.6 Conversation-Level Sentiment Shifts

I analyzed 200 multi-turn conversations and tracked sentiment changes from start to finish. Around two-thirds displayed a shift toward more positive sentiment, even if the change was modest. This suggests the model gently nudges the conversation in a more reassuring direction without resorting to forced positivity.

Some conversations remained emotionally flat. A few even declined, usually because the user continued to express frustration or sadness. This reminds us that chatbot responses cannot magically resolve emotional struggles. At best, they provide a small sense of companionship.

$$\text{Sentiment}(x) = \sum_{i=1}^n w_i \cdot s_i$$

8.7 Comparison with Baselines

Compared with a simple keyword-matching system:

- emotion classification improved dramatically,
- responses became more fluid and context-sensitive,
- empathy cues increased significantly,
- and conversational warmth rose across nearly all categories.

The baseline system often sounded scripted. EmoCare, by contrast, felt more like a patient listener. Even when its responses were not perfect, they were rarely tone-deaf. This alone marks a substantial improvement.

9 Discussion

Building an emotionally attentive chatbot inevitably raises questions about what “good” emotional support even looks like. The results suggest that fine-grained classification combined with a small, controlled generator produces interactions that feel more sincere than template-driven systems. Yet sincerity in a machine is an illusion. The responses sound warm because patterns in the training data encourage that tone, not because the model understands suffering or relief.

9.1 Interpreting the Results

The improvements across classification and generation indicate progress, but not a solution. A model may recognize embarrassment or pride with moderate accuracy, but emotional meaning is deeply shaped by culture, personality, and context. What feels supportive in one setting might feel intrusive in another.

Despite this, the idea of creating systems that simply respond a bit more thoughtfully has value. Many people use chatbots as a first step toward acknowledging uncomfortable feelings. If software can make that experience gentler without pretending to offer therapy, it may serve a real purpose.

9.2 Strengths of the Approach

Several aspects stand out:

- Using thirty-two emotions forces the classifier to pay attention to subtle cues.
- Semi-supervised learning helps counter data imbalance without requiring new annotations.
- A small generator avoids the overly polished, sometimes uncanny tone of large models.
- Psycholinguistic evaluation provides a better picture of emotional resonance.

Taken together, these components form a balanced pipeline.

9.3 Limitations

There are several limitations worth acknowledging. First, the dataset, while thoughtfully designed, does not reflect the full complexity of real-world emotional conversations. People speak differently when they trust someone deeply, when they are under stress, or when they are hiding something. None of this richness appears consistently in the data.

Second, language models, even small ones, occasionally produce responses that miss the mark emotionally. They might offer reassurance when silence would be more empathetic, or ask a question that feels unnecessary.

Finally, evaluating empathy remains difficult. Metrics cannot capture how a message makes someone feel. Real validation requires human feedback, ideally from diverse cultural backgrounds.

10 Ethical Considerations

Working with emotional dialogue demands caution. A chatbot, no matter how well-designed, cannot replace professional care. Users may mistakenly rely on it during crises. To address this, the system avoids making strong claims, offering diagnoses, or giving explicit advice.

10.1 Data Privacy

Emotional data is sensitive by nature. Conversations must be stored securely, anonymized when possible, and used responsibly. Transparency about data usage is essential. Users deserve to know how their messages are handled.

10.2 Bias and Fairness

Models inevitably reflect the biases present in their training data. If certain emotional expressions occur more commonly in particular demographic groups, the classifier may skew toward certain interpretations. Careful auditing is necessary to ensure the system does not misinterpret emotions due to cultural differences.

10.3 Appropriate Use

This system is designed as a supportive companion, not as a therapist. It should be integrated into environments where human support is available when needed. Automated escalation mechanisms may be useful for detecting crisis language.

11 Conclusion

EmoCare explores whether emotionally sensitive chatbots can feel more grounded without relying on massive generative models. Through a combination of fine-grained emotional classification, semi-supervised learning, and a controlled LLaMA-style generator, the system produces responses that sound more attentive and less mechanical than common baseline approaches.

While the work does not claim to replicate human warmth, it does show that careful modeling choices can make chatbot interactions feel more natural. There remains a long road ahead in understanding emotional communication between humans and machines. This project serves as one step in that direction, suggesting that small adjustments in emotional awareness can influence not just performance metrics but the overall tone of conversation.

References

- [1] Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. *JMIR Mental Health*, 4(2), e19.
- [2] Evaluating the Effectiveness of Artificial Intelligence–Driven Chatbots in Mental Health Support: Systematic Review. *Journal of Medical Internet Research*, 2023. Available at: <https://www.jmir.org/2023/1/e51712>
- [3] The Use of Mobile Health Applications for Mental Health Interventions: Progress and Challenges. *JMIR mHealth and uHealth*, 2018. Available at: <https://mhealth.jmir.org/2018/11/e12106/>
- [4] AI-Based Conversational Agents in Healthcare: A Review of Emerging Diagnostic and Support Systems. *JMIR Medical Informatics*, 2025. Available at: <https://medinform.jmir.org/2025/1/e63538>
- [5] Advanced Emotional Support Chatbots: Linguistic and ML-Based Approaches. Google Research Document, 2024. Available at: <https://share.google/iaelqfVNOwehatbgQ>
- [6] Rashkin, H., Smith, E. M., Li, M., & Boureau, Y.-L. (2019). Towards Empathetic Open-domain Conversation Models: A New Benchmark and Dataset. In *Proceedings of ACL* (pp. 5370–5381).
- [7] Zhou, H., Huang, M., Zhang, T., Zhu, X., & Liu, B. (2018). Emotional Chatting Machine: Emotional Conversation Generation with Internal and External Memory. In *Proceedings of AAAI*.
- [8] Sun, Y., Lin, J., & Wen, X. (2021). Empathetic Dialogue Generation: A Review of Datasets, Methods, and Evaluation Metrics. *IEEE Transactions on Affective Computing*.
- [9] Lubis, N., Ultes, S., Mairesse, F., Engelke, D., Rashed, A., et al. (2018). Eliciting User Emotional State Using Generative Spoken Dialogue Models. In *Proceedings of SIGDIAL*.

- [10] Ghosh, S., et al. (2021). Affective Dialogue Generation with Transformer-Based Models. In *Proceedings of EMNLP Workshops*.
- [11] Lin, W., & Xu, Z. (2020). The BIRDS Framework: Integrating Emotion, Empathy, and Personalization in Chatbot Design. In *Proceedings of CHI*.
- [12] Ma, X., & Chen, J. (2022). AI Companions for Mental Health Support: Psychology-Informed Design and Linguistic Analysis. *Nature Digital Medicine*.
- [13] Vlasov, V. (2018). FEELU: A Rule-Based and Machine-Learning Enhanced Empathetic Dialogue System. *arXiv preprint arXiv:1811.00207*.